FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

FLAB3

Cayanan, Nathaniel C.	Tejuco, Hadji J.
Name of Student	Name of Professor
10-29-2026	10-29-2026
Data Performed	Date Submitted

MIDTERM EXAM

FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

List the steps that you perform to come up with the program.

1. Asks the user to type in all their personal school information.
2. Displays a clean receipt showing their info and the hardcoded list of subjects.
3. Prompts the user to type in their fees one by one.
4. Adds all the numbers together, translates the total, prints the final amount, and safely closes the program.

Paste the source code and explain the functionalities and take a screenshot of the final output.

Enrollment Status : ENROLLED
Student # : 202511057
Name : CAYANAN, NATHANIEL CLIFFORD CRAFT
Phone : +639273736323 Zip Code : 1940
Classification :

College : COMPUTER STUDIES Program : BSCSSE
Year Level : 2 1st term, SY 26-27
Address : west drive

Courses	Title	Section	Units	Days	Time	Room
CCS0021	INFORMATION MANAGEMENT (LEC)	TN24	2	T	16:00:00-18:40:00	ONLINE
CCS0021L	INFORMATION MANAGEMENT (LAB)	TN24	1	F	16:00:00-18:50:00	ONLINE
CS0001	DISCRETE STRUCTURES 1	TN24	3	S / W	15:00:00-16:50:00 / 15:00:00-16:50:00	F710 / F710
CS0003	COMPUTER SYSTEMS & ARCHITECTURE	TN24	2	F	10:00:00-12:40:00	ONLINE
CS0003L	COMPUTER SYSTEMS & ARCHITECTURE (LAB)	TN24	1	TH	10:00:00-12:50:00	E610
CS0070	OBJECT ORIENTED PROGRAMMING	TN24	2	F	07:00:00-09:40:00	ONLINE
CS0070L	OBJECT ORIENTED PROGRAMMING-LAB	TN24	1	TH	07:00:00-09:50:00	E609
GED0021	SPECIALIZED ENGLISH PROGRAM 2	TN24	3	S / W	11:00:00-12:50:00 / 11:00:00-12:50:00	F607 / F607
GED0075	LINEAR ALGEBRA	TN24	3	F / T	13:00:00-14:50:00 / 13:00:00-14:50:00	ONLINE / ONLINE
GED0081	COLLEGE PHYSICS 1 LECTURE	TN24	2	T	07:00:00-09:40:00	ONLINE
GED0081L	COLLEGE PHYSICS 1 LABORATORY	TN24	1	S	07:00:00-09:50:00	F1003
TOTAL UNITS 21						

Tuition Fee (21)	44,268.00
Miscellaneous Fee	12,305.00
ID Validation Fee	80.00
ITE Computer Laboratory Fee (CCS0021)	4,008.00
ITE Computer Laboratory Fee (CS0003)	4,008.00
ITE Computer Laboratory Fee (CS0070)	4,008.00
Science Lab Fee (GED0081)	4,104.00
TOTAL ASSESSMENT	72,781.00
Down Payment (Upon Enrollment) 40%	29,112.40
Midterm (Oct 05 - Oct 07 2026) 30%	21,834.30
Final (Nov 26 - Dec 03 2026) 30%	21,834.30

onlinegdb.com/online_nasm_assembler

OnlineGDB
online compiler and debugger for c/c++

code. compile. run. debug. share.

IDE
My Projects
Classroom new
Learn Programming
Programming Questions
Sign Up
Login

wegic

Modern, dynamic and visually stunning websites created from a simple business description.

ADS VIA BUYSELLADS


```
Enter Enrollment Status: ENROLLED
Enter Student #: 202511057
Enter Name: CAYANAN, NATHANIEL CLIFFORD CRAFT
Enter Phone: 09273736323
Enter Zip Code: 1940
Enter College: COMPUTER STUDIES
Enter Program: BSCSSE
Enter Year Level: 2
Enter Term: 1
Enter School Year (SY): 2026-2027
Enter Address: west drive

================================================
                REGISTRATION SUMMARY
================================================
Enrollment Status: ENROLLED
Student #        : 202511057
Name             : CAYANAN, NATHANIEL CLIFFORD CRAFT
Phone            : 09273736323
Zip Code         : 1940
College          : COMPUTER STUDIES
Program          : BSCSSE
Year Level       : 2
Term             : 1
School Year (SY) : 2026-2027
Address          : west drive
Class Section    : TN24

Enrolled Courses:
  - CCS0021   INFORMATION MANAGEMENT (LEC)
  - CCS0021L  INFORMATION MANAGEMENT (LAB)
  - CS0001    DISCRETE STRUCTURES 1
  - CS0003    COMPUTER SYSTEMS & ARCHITECTURE
  - CS0003L   COMPUTER SYSTEMS & ARCHITECTURE (LAB)
  - CS0070    OBJECT ORIENTED PROGRAMMING
  - CS0070L   OBJECT ORIENTED PROGRAMMING-LAB
  - GED0021   SPECIALIZED ENGLISH PROGRAM 2
  - GED0070   LINEAR ALGEBRA
  - GED0081   COLLEGE PHYSICS 1 LECTURE
  - GED0081L  COLLEGE PHYSICS 1 LABORATORY
TOTAL UNITS: 21


================================================
                FINANCIAL ASSESSMENT
================================================
Enter Tuition Fee Amount: 44268
Enter Miscellaneous Fee Amount: 12305
Enter Lab Fees Amount: 16128

TOTAL ASSESSMENT VALUE: Php 72701


...Program finished with exit code 0
Press ENTER to exit console.
```


About • FAQ • Docs • Blog • Terms of Use • Contact Us • GDB Tutorial • Credits • Privacy

```
section .data
    prompt_status   db "Enter Enrollment Status: ", 0
    prompt_id       db "Enter Student #: ", 0
    prompt_name     db "Enter Name: ", 0
    prompt_phone    db "Enter Phone: ", 0
    prompt_zip      db "Enter Zip Code: ", 0
    prompt_college  db "Enter College: ", 0
    prompt_program  db "Enter Program: ", 0
    prompt_year     db "Enter Year Level: ", 0
    prompt_term     db "Enter Term: ", 0
    prompt_sy       db "Enter School Year (SY): ", 0
    prompt_address  db "Enter Address: ", 0

    prompt_tuition  db "Enter Tuition Fee Amount: ", 0
    prompt_misc     db "Enter Miscellaneous Fee Amount: ", 0
    prompt_lab      db "Enter Lab Fees Amount: ", 0

    hdr_info        db 10,
"=====================================================", 10, \
                    "             REGISTRATION SUMMARY              ", 10, \

"=====================================================", 10, 0
    lbl_status      db "Enrollment Status: ", 0
    lbl_id          db "Student #        : ", 0
    lbl_name        db "Name             : ", 0
    lbl_phone       db "Phone            : ", 0
    lbl_zip         db "Zip Code         : ", 0
    lbl_college     db "College          : ", 0
    lbl_program     db "Program          : ", 0
    lbl_year        db "Year Level       : ", 0
    lbl_term        db "Term             : ", 0
    lbl_sy          db "School Year (SY) : ", 0
    lbl_address     db "Address          : ", 0
    lbl_section     db "Class Section    : TN24", 10, 0

    hdr_courses     db 10, "Enrolled Courses:", 10, \
                    "  - CCS0021   INFORMATION MANAGEMENT (LEC)", 10, \
                    "  - CCS0021L  INFORMATION MANAGEMENT (LAB)", 10, \
                    "  - CS0001    DISCRETE STRUCTURES 1", 10, \
                    "  - CS0003    COMPUTER SYSTEMS & ARCHITECTURE", 10, \
                    "  - CS0003L   COMPUTER SYSTEMS & ARCHITECTURE (LAB)",
10, \
                    "  - CS0070    OBJECT ORIENTED PROGRAMMING", 10, \
                    "  - CS0070L   OBJECT ORIENTED PROGRAMMING-LAB", 10, \
                    "  - GED0021   SPECIALIZED ENGLISH PROGRAM 2", 10, \
                    "  - GED0070   LINEAR ALGEBRA", 10, \
                    "  - GED0081   COLLEGE PHYSICS 1 LECTURE", 10, \
```

```
                    "  - GED0081L  COLLEGE PHYSICS 1 LABORATORY", 10, \
                    "TOTAL UNITS: 21", 10, 0

    hdr_billing     db 10,
"=====================================================", 10, \
                    "              FINANCIAL ASSESSMENT              ", 10, \

"=====================================================", 10, 0
    lbl_total       db "TOTAL ASSESSMENT VALUE: Php ", 0
    newline         db 10, 0

section .bss
    buffer_status   resb 32
    buffer_id       resb 32
    buffer_name     resb 64
    buffer_phone    resb 32
    buffer_zip      resb 16
    buffer_college  resb 64
    buffer_program  resb 32
    buffer_year     resb 16
    buffer_term     resb 32
    buffer_sy       resb 32
    buffer_address  resb 128

    buffer_money    resb 32
    running_total   resd 1
    string_out      resb 32

section .text
    global _start

%macro OUTPUT_TEXT 1
    mov ecx, %1
    xor edx, edx
%%count_bytes:
    cmp byte [ecx + edx], 0
    je %%write_syscall
    inc edx
    jmp %%count_bytes
%%write_syscall:
    mov eax, 4
    mov ebx, 1
    int 0x80
%endmacro

%macro INPUT_TEXT 2
    mov eax, 3
    mov ebx, 0
    mov ecx, %1
```

```
    mov edx, %2
    int 0x80

    xor ecx, ecx
%%clean_newline:
    cmp ecx, eax
    jge %%exit_macro
    mov edx, %1
    cmp byte [edx + ecx], 10
    jne %%inc_pointer
    mov byte [edx + ecx], 0
%%inc_pointer:
    inc ecx
    jmp %%clean_newline
%%exit_macro:
%endmacro

%macro DISPATCH_PROMPT 3
    OUTPUT_TEXT %1
    INPUT_TEXT %2, %3
%endmacro

_start:
    DISPATCH_PROMPT prompt_status, buffer_status, 32
    DISPATCH_PROMPT prompt_id, buffer_id, 32
    DISPATCH_PROMPT prompt_name, buffer_name, 64
    DISPATCH_PROMPT prompt_phone, buffer_phone, 32
    DISPATCH_PROMPT prompt_zip, buffer_zip, 16
    DISPATCH_PROMPT prompt_college, buffer_college, 64
    DISPATCH_PROMPT prompt_program, buffer_program, 32
    DISPATCH_PROMPT prompt_year, buffer_year, 16
    DISPATCH_PROMPT prompt_term, buffer_term, 32
    DISPATCH_PROMPT prompt_sy, buffer_sy, 32
    DISPATCH_PROMPT prompt_address, buffer_address, 128

    OUTPUT_TEXT hdr_info
    OUTPUT_TEXT lbl_status
    OUTPUT_TEXT buffer_status
    OUTPUT_TEXT newline

    OUTPUT_TEXT lbl_id
    OUTPUT_TEXT buffer_id
    OUTPUT_TEXT newline

    OUTPUT_TEXT lbl_name
    OUTPUT_TEXT buffer_name
    OUTPUT_TEXT newline

    OUTPUT_TEXT lbl_phone
```

```
OUTPUT_TEXT buffer_phone
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_zip
OUTPUT_TEXT buffer_zip
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_college
OUTPUT_TEXT buffer_college
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_program
OUTPUT_TEXT buffer_program
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_year
OUTPUT_TEXT buffer_year
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_term
OUTPUT_TEXT buffer_term
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_sy
OUTPUT_TEXT buffer_sy
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_address
OUTPUT_TEXT buffer_address
OUTPUT_TEXT newline

OUTPUT_TEXT lbl_section
OUTPUT_TEXT hdr_courses
OUTPUT_TEXT newline

OUTPUT_TEXT hdr_billing
mov dword [running_total], 0

DISPATCH_PROMPT prompt_tuition, buffer_money, 32
call procedure_ascii_to_int
add [running_total], eax

DISPATCH_PROMPT prompt_misc, buffer_money, 32
call procedure_ascii_to_int
add [running_total], eax

DISPATCH_PROMPT prompt_lab, buffer_money, 32
call procedure_ascii_to_int
add [running_total], eax
```

```
    OUTPUT_TEXT newline
    OUTPUT_TEXT lbl_total

    mov eax, [running_total]
    call procedure_int_to_ascii
    OUTPUT_TEXT string_out
    OUTPUT_TEXT newline
    OUTPUT_TEXT newline

    mov eax, 1
    mov ebx, 0
    int 0x80

procedure_ascii_to_int:
    xor eax, eax
    xor ecx, ecx
    mov esi, buffer_money
.read_character:
    movzx edx, byte [esi + ecx]
    cmp dl, 0
    je .execution_finished
    cmp dl, 10
    je .execution_finished

    cmp dl, '0'
    jl .skip_character
    cmp dl, '9'
    jg .skip_character

    sub dl, '0'
    imul eax, 10
    add eax, edx
.skip_character:
    inc ecx
    jmp .read_character
.execution_finished:
    ret

procedure_int_to_ascii:
    mov ebx, 10
    mov edi, string_out + 31
    mov byte [edi], 0
.division_loop:
    xor edx, edx
    div ebx
    add dl, '0'
    dec edi
    mov [edi], dl
```

```
    test eax, eax
    jnz .division_loop

    mov esi, edi
    mov edi, string_out
.string_alignment_copy:
    lodsb
    stosb
    test al, al
    jnz .string_alignment_copy
    ret
```